

Foucault Networks on Celestial Bodies

Frame-consistent local dynamics and nested observation maps

Stas

Independent Research Program; ORCID: 0009-0000-2294-705X

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Status. This document supersedes the bilingual working draft v1.0. It is migrated to GO Core v0.2 and the Frames-Forces-Dissipation Interface v0.1. The local pendulum is a small-amplitude rotating-frame model; the celestial hierarchy is a finite kinematic laboratory, not an ephemeris or a precision geophysical model.

Abstract

We formulate a network of idealized Foucault pendula on a finite hierarchy of rotating bodies without mixing body, local, inertial, and observer-frame components. A body edge in the hierarchy is transformed through its parent's attitude, a local endpoint displacement is transformed together with the site position, and a moving observer is represented by a full affine frame map rather than an origin subtraction alone. In the body-fixed tangent plane the declared small-amplitude model is

$$\ddot{u} + 2\beta\dot{u} + 2\Omega_n J\dot{u} + \omega_0^2 u = 0, \quad \Omega_n = \Omega^B \cdot n^B.$$

For $\beta = 0$, the substitution $u_C = e^{-i\Omega_n t} u$ proves that the oscillation plane precesses at signed rate $-\Omega_n$ in the chosen orientation. The Coriolis term performs no work, while damping dissipates $P_{\text{diss}} = 2m\beta \|\dot{u}\|^2 \geq 0$. A nested quasiperiodic hierarchy is represented as a continuous readout of a torus flow. Its closure is the image of the subtorus determined by integer frequency relations. Frequency support is preserved under a static Euclidean change of observer and a time-origin shift, but not under an arbitrary time-dependent rotating observer or a lossy camera projection. The result is a typed observation model with explicit claim boundaries, not a new theory of celestial mechanics.

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1 Scope and observation protocol

The experiment is a finite mathematical laboratory: choose bodies b , surface sites ℓ , local pendula, an observer frame O , and a sampling/camera protocol λ . The typed chain is

$$X_t \xrightarrow{\mathcal{R}} \tilde{X}_t \xrightarrow{\mathcal{O}_\lambda} Z_t \xrightarrow{\mathcal{K}_\lambda} \text{Law}(Y_t | Z_t) \xrightarrow{\hat{\theta}_\lambda} \hat{\theta}_t.$$

The reduction $\mathcal{R}$ selects a finite hierarchy and local model, $\mathcal{O}_\lambda$ produces ideal framed endpoints or projected pixels, $\mathcal{K}_\lambda$ represents sampling and noise, and $\hat{\theta}_\lambda$ estimates rates or spectral relations. These maps are not identified.

Boundary 1.1 (Claim firewall). The model excludes tidal elasticity, polar motion, atmospheric loading, suspension anisotropy, finite-amplitude precession, general relativity, and precision ephemerides. Parameters called “planetary” or “galactic” are labels unless supplied by a dated external model with uncertainty.

2 Frame-consistent celestial hierarchy

Fix an inertial reference I . Each body b has a center $R_b^I(t)$ and attitude

$$A_{I \leftarrow B_b}(t) \in \text{SO}(3),$$

where B_b is its body frame. If $p(b)$ is the parent and the relative center vector is stored in the parent frame, recursion is

$$R_b^I = R_{p(b)}^I + A_{I \leftarrow B_{p(b)}} r_{b|p(b)}^{B_{p(b)}}. \quad (1)$$

Thus every addition in equation (1) occurs in I .

At site ℓ , let $r_{b\ell}^{B_b}$ be the body-fixed site point, $n_{b\ell}^{B_b}$ the outward unit normal, and

$$E_{B_b \leftarrow L_{b\ell}} = \begin{pmatrix} e_1^{B_b} & e_2^{B_b} & n^{B_b} \end{pmatrix} \in \text{SO}(3)$$

the local frame. A horizontal pendulum displacement is $u^{L_{b\ell}} = (u_1, u_2, 0)^\top$.

Definition 2.1 (Endpoint point map). The endpoint in the inertial frame is

$$p_{b\ell}^I = R_b^I + A_{I \leftarrow B_b} \left(r_{b\ell}^{B_b} + E_{B_b \leftarrow L_{b\ell}} u^{L_{b\ell}} \right). \quad (2)$$

For a moving observer with origin $c_O^I(t)$ and attitude $A_{I \leftarrow O}(t)$, the observer components are

$$p_{b\ell}^O = A_{O \leftarrow I} (p_{b\ell}^I - c_O^I). \quad (3)$$

A camera map Π_λ then gives $z_{b\ell} = \Pi_\lambda(p_{b\ell}^O)$. In general Π_λ is not invertible.

Remark 2.2 (Correction to v1.0). The expression $R_b^I + A_{I \leftarrow B_b} r_{b\ell}^{B_b} + u_{b\ell}^{B_b} - c_O^I$ mixed the body displacement $u_{b\ell}^{B_b}$ with inertial vectors. Equation equation (2) transforms the site and local displacement before their inertial addition; equation (3) also includes observer attitude.

3 Local rotating-frame pendulum

Let the body angular velocity be Ω^B and set

$$\Omega_n = \Omega^B \cdot n^B.$$

On the oriented tangent plane define

$$Jv = n^B \times v, \quad J^2 = -I.$$

The following is a declared reduced model, not the exact spherical pendulum.

Model 3.1 (Linear Foucault layer). For pendulum length $\ell_p > 0$, $\omega_0^2 = g_{\text{eff}}/\ell_p$, and damping rate $\beta \geq 0$, the body-fixed horizontal displacement obeys

$$\ddot{u} + 2\beta\dot{u} + 2\Omega_n J\dot{u} + \omega_0^2 u = 0. \quad (4)$$

The model assumes a fixed local vertical, small displacement, constant Ω_n , and neglects terms beyond the horizontal Coriolis projection.

Theorem 3.2 (Signed precession in the undamped reduced model). *Under model 3.1 with $\beta = 0$, identify the oriented tangent plane with $\mathbb{C}$, so J acts as multiplication by i . Every solution has the form*

$$z(t) = e^{-i\Omega_n t} (C_+ \cos(\omega_c t) + C_- \sin(\omega_c t)), \quad \omega_c = \sqrt{\omega_0^2 + \Omega_n^2}.$$

Consequently a linearly polarized oscillation plane precesses in the body frame at signed rate

$$\dot{\alpha} = -\Omega_n$$

for the stated orientation convention.

Proof. With $z = e^{-i\Omega_n t} w$, the complex form

$$\ddot{z} + 2i\Omega_n \dot{z} + \omega_0^2 z = 0$$

reduces to $\ddot{w} + (\omega_0^2 + \Omega_n^2)w = 0$. □

For a spherical body whose spin axis defines latitude λ ,

$$\Omega_n = \Omega \sin \lambda.$$

Hence the magnitude vanishes at the equator and equals $|\Omega|$ at the poles. The sign depends on the orientation of n, e_1, e_2 and on the sign of the body's rotation; only after these are declared is “clockwise” meaningful.

4 Dissipation and energy

Define the reduced mechanical energy

$$E_{\text{loc}} = \frac{m}{2} \|\dot{u}\|^2 + \frac{m\omega_0^2}{2} \|u\|^2.$$

Proposition 4.1 (Coriolis work and damping sign). *Every solution of equation (4) satisfies*

$$\dot{E}_{\text{loc}} = -2m\beta \|\dot{u}\|^2. \quad (5)$$

Thus the Coriolis term performs no work and

$$P_{\text{diss}} = 2m\beta \|\dot{u}\|^2 \geq 0.$$

Proof. Dot equation (4) with $m\dot{u}$. Since $\dot{u} \cdot J\dot{u} = 0$, the Coriolis contribution vanishes. $\square$

Equation (5) does not imply that the full body-pendulum-support energy is autonomous. Body rotation and suspension control are externalized by the reduced model.

5 Nested torus layer

Let the active phases be

$$\Theta(t) = \Theta_0 + \omega t \pmod{2\pi}, \quad \Theta \in \mathbb{T}^m.$$

A finite circular hierarchy and local oscillation can be represented by a continuous map $F : \mathbb{T}^m \rightarrow \mathbb{R}^3$, with ideal trace

$$Y(t) = F(\Theta(t)).$$

This includes finite Fourier readouts

$$Y(t) = \sum_{k \in \mathcal{K}} c_k e^{i(k \cdot \omega)t}, \quad \mathcal{K} \subset \mathbb{Z}^m \text{ finite.}$$

Theorem 5.1 (Orbit closure of the phase flow). *Let*

$$\Lambda_\omega = \{k \in \mathbb{Z}^m : k \cdot \omega = 0\}.$$

The closure of $\Theta(t)$ is the translate by Θ_0 of the closed subtorus whose annihilator is Λ_ω . In particular:

- (i) *the flow is periodic iff some $T > 0$ satisfies $T\omega \in 2\pi\mathbb{Z}^m$;*
- (ii) *if the components of ω are rationally independent, the phase orbit is dense in $\mathbb{T}^m$;*
- (iii) $\text{cl}\{Y(t)\} = F(\text{cl}\{\Theta(t)\})$.

Proof. The first two statements are the standard character description of a one-parameter subgroup of a torus. The last follows because the phase closure is compact and F is continuous. $\square$

The map F can collapse dimensions or frequency amplitudes. Therefore a full phase torus need not be identifiable from one projected trace.

6 Transformation audit

Quantity or structure	Declared transformation	Status
$\Omega^B \cdot n^B$	common $A \in \text{SO}(3)$ applied to both vectors	invariant scalar
Endpoint p^I	affine Euclidean frame change	covariant point, not invariant components

Quantity or structure	Declared transformation	Status
Local energy E_{loc}	static tangent-plane orthogonal basis change	invariant
Orbit-closure topology	static observer-space homeomorphism	preserved
Frequency set	static Euclidean isometry and time-origin shift	preserved up to zero/trend conventions
Frequency amplitudes	camera projection	may be suppressed
Frequency set	arbitrary time-dependent rotating observer	not invariant; modulation and shifts may occur
Signed precession	reversal of tangent-plane orientation	changes sign

Boundary 6.1 (No universal spectral invariant). “Frequency support” is not an invariant under every observer choice. The admissible statement requires a fixed clock and a static Euclidean change of observer, or a specified correction for the observer motion.

7 Non-identifiability

Proposition 7.1 (One trace does not identify one hierarchy). *For a single linear camera projection and finite observation horizon, the map from hierarchy parameters, local site state, and phase amplitudes to the projected trace is not injective in general.*

Proof. Any component in the kernel of the camera projection is invisible. Furthermore, two latent Fourier channels whose projected coefficient sum is equal produce the same projected signal when their frequencies coincide. Finite sampled data additionally admit aliases separated by integer multiples of the sampling angular frequency. Hence distinct hidden records can have the same observation. $\square$

Recoverability requires extra hypotheses: multiple noncoplanar camera channels, multiple sites, known body attitudes, anti-alias sampling, or external ephemeris constraints. The present document proves none of these sufficient in a realistic network.

8 Analytic sanity checks

Using the conventional nominal terrestrial rotation magnitude $\Omega_{\oplus} = 7.2921150 \times 10^{-5} \text{ s}^{-1}$, the reduced-model precession period is

$$T_F(\lambda) = \frac{2\pi}{\Omega_{\oplus} |\sin \lambda|}$$

when $\sin \lambda \neq 0$. This gives:

Latitude	$\Omega_n/\Omega_\oplus$	T_F
0°	0	no reduced-model precession
45°	$1/\sqrt{2}$	33.85 h
90°	1	23.93 h

These are formula checks, not predictions for a real installation. The nominal rotation value does not encode polar motion or length-of-day variations.

9 Claim boundary

The strict contributions are frame correction, typed observation maps, a proved precession statement inside a declared reduced model, and qualified transformation laws. This document does not claim:

- a new derivation of the Foucault effect;
- a KAM theorem for the Solar System;
- a realistic pendulum on another celestial body;
- recovery of a celestial hierarchy from one trace;
- an empirical test or a new physical parameter.

References

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